

A Machine-Checked Cutoff Theorem for the Census of an Exhaustive Combinatorial-Optimality Search

Abstract

Exhaustive searches for combinatorial optima are re-run from scratch at every instance size, and their cost is dominated by the set of states they store. We identify a class of such searches — *graded ambient searches* — for which that stored set stabilises: for every level ℓ and every ambient n above a cutoff $n_{\min}(\ell)$, the census at n is determined by one computation at $n_{\min}(\ell)$ together with an explicit *affine* growth term of slope exactly one. The theorem, its level-agreement clause, and a conditional polynomial refinement are machine-checked in Isabelle/HOL with two non-vacuous interpretations. The closest prior art is dynamic cutoff detection for parameterized concurrent programs; we state what survives the comparison. We instantiate the class in three domains, use it to re-certify $S(11) = 35$ against an unmodified verified checker, and report the two measured walls at which the method’s practical reach ends.

1 Introduction

An exhaustive search for a combinatorial optimum spends its budget on *storage*. Branch-and-bound with memoisation, dynamic programming over canonical states, and the depth- or size-optimality searches for sorting networks [30, 8, 11, 24] all share the same shape: a root state, a transition relation, a bound oracle, and a memo table whose size is the binding constraint. The memo table is a function of the instance parameter n , and the conventional assumption is that each n demands its own computation. When the search is already at the edge of a machine at n , the computation at $n + 1$ is simply out of reach.

This paper identifies a structural condition under which that assumption is false. Let a search carry, besides its instance parameter n — which we call the *ambient* — an intrinsic *width* $w(s) \leq n$ on states, and let its cost budget be reparameterised as a *level* ℓ above a forced prefix of the search whose cost $C(n)$ is known in closed form. Write $\text{Reach}(n, \ell)$ for the census: the set of states the search stores at ambient n and level ℓ . Under five hypotheses on naming, transitions, grading and geodesics, the census satisfies an exact identity

$$\text{Reach}(n, \ell) = \text{Reach}(n_{\min}(\ell), \ell) \uplus \{\text{root}_w : n_{\min}(\ell) < w \leq n\},$$

so that its cardinality grows by exactly one per unit of ambient above the cutoff. One computation determines the census at every larger instance size, forever. We prove this in two logically independent halves — an *ambient collapse* that needs no cutoff at all, and a *chain collapse* that supplies one — and we machine-check both.

Prior art, conceded first. This is a *cutoff theorem*, and cutoffs are a mature named concept in parameterized verification [19, 18, 31, 1, 28, 25, 5, 6]. The closest prior art is not a distant analogy but a direct one. Kaiser, Kroening and Wahl [29], Definition 4, define a cutoff as “a number $c \in \mathbb{N}$ such that, for all $n \geq c$, $R_n = R_c$ ”, where R_n is a *set of reachable states* of an n -thread program — not the truth value of a property. An earlier version of the present work claimed to be the first cutoff theorem about a set rather than a property. That claim is false

and we retract it here. Three distinctions survive, and we claim only these. (i) R_n in [29] is a set of *abstract* thread states under a fixed finite abstraction; our census is the concrete set of states an exhaustive optimality search actually stores, at full fidelity. (ii) [29] concludes exact equality with no growth term; we give an affine growth law with slope exactly one, and, conditionally, a polynomial one. (iii) Their cutoff is *dynamically detected* by a sufficient runtime test during the analysis; ours is an *a-priori* consequence of the hypotheses, computable before the search is launched. Two adjacent bodies of work are not prior art: symmetry and partial-order reduction shrink an explored state space by a quotient or a representative selection [10, 27, 34], and work on models of dynamic programming and dominance caching in branch-and-bound [9, 13] concerns expressiveness and pruning — neither asserts that the object explored at one parameter equals the object explored at another. We use “cutoff” in the tight-cutoff sense of Emerson–Namjoshi [19, 20] and Außerlechner–Jacobs–Khalimov [1], with the census of the explored state set in place of the truth of a temporal property; not in the Emerson–Kahlon sense [18]. Namjoshi’s completeness result for invariance [31] is the other landmark of the a-priori side: there the cutoff exists but finding it is undecidable, whereas ours is read off the hypotheses.

Scope, conceded second. The class defines the census as a ball in a level-independent weighted distance. The engine we instantiate it on stores a *bound-gated subset* of that ball, because its recursion reaches successor expansion only when a Huffman-style bound test passes. The two disagree already at level 1. The consequence is stated once and enforced throughout: the sorting-network identity is proved directly from the gate, not as a corollary of the abstract theorem, and the claim that the sorting-network search satisfies the five hypotheses is *not* machine-checked. It rests on an exhaustive audit of the implementation, reported as such.

Origin. The class was extracted from a concrete question. A 1972 lower-bound argument for sorting networks is the sole published support for a value of $S(13)$ that circulates in the two most-consulted public tables, and a companion paper [2] shows that the argument does not establish it. Auditing the computational route to a replacement meant asking what a size-optimality search at 13 channels would have to store, which meant relating it to the same search at 11 channels; that is where the collapse identity came from. The audit does not recur here beyond Section 5, where the sorting-network search is one of three instances.

Road map. Section 2 defines the class and states the five hypotheses, including the one — width convexity — that was invisible in the concrete statement because it happens to be free there. Section 3 states both halves of the theorem, the level-agreement clause and the growth law, separating the affine law we prove from the polynomial law we do not. Section 4 reports the Isabelle development, the two interpretations that make it non-vacuous, three corrections the mechanization forced on our own hypotheses, and what is not mechanized. Sections 5 and 6 give three instances; Section 7 the certified computation; Section 8 the measured boundary; Section 9 the trusted base. Everything a reader must audit is in the paper; the theory file, the engine, the certificate and the verifiers are in the deposit [3].

2 Graded ambient searches

Definition 1 (graded ambient search). A *graded ambient search* is a tuple $(S, w, \rightarrow, \gamma, \{\text{root}_n\})$ in which S is a set of canonical states, $w : S \rightarrow \mathbb{N}$ assigns each state an intrinsic *width*, $\rightarrow_n \subseteq S \times S$ is a transition relation for each ambient n , γ assigns each edge a non-negative integer *grade*, and $\text{root}_n \in S$ is the ambient- n root. Crucially S itself does not depend on n . Writing d_n for the γ -weighted distance in $\rightarrow_n$ and $C(n)$ for the free-chain offset of H3 below, the object of study is the *census*

$$\text{Reach}(n, \ell) = \{s \in S : d_n(\text{root}_n, s) \leq C(n) + \ell\},$$

where ℓ is the *level* and, for a width value v , $W(v, \ell) = \{s \in \text{Reach}(v, \ell) : w(s) = v\}$ is the width- v *stratum*.

Three features are deliberate. The census is the set of states an exhaustive search *stores*, not the abstract solution set: it is what costs money. The width is intrinsic to the state, not to the run that found it, so canonicalisation must quotient away unused dimensions. And the budget is reparameterised: a search whose first $C(n)$ units of progress are forced is studied at level $\ell = L - C(n)$, not at total cost L . Without that shift the ambient dependence sits entirely in the offset and no identity is visible.

2.1 The hypotheses

H0 (ambient-free naming). The canonical name of a state is a function of the state alone; canonicalisation quotients away unused dimensions, so a state has one and the same name at every ambient $n \geq w(s)$.

H1 (ambient freedom, stratified). For every s and all $w(s) \leq m \leq n$, $\{t : s \rightarrow_n t, w(t) \leq m\} = \{t : s \rightarrow_m t\}$. Raising the ambient may only add transitions to *wider* states.

H2 (ambient-free grading). $\gamma(s \rightarrow t)$ is a function of (s, t) alone.

H3 (root tower). $\text{root}_n \rightarrow_n \text{root}_{n-1}$ with grade $\delta(n)$, and $C(n) = \sum_{k \leq n} \delta(k)$ is the *free chain*. The *degenerate* case $\text{root}_n = \text{root}_0$, $C \equiv 0$, is explicitly allowed.

H4 (width convexity). For all $m \leq n$ and every s with $w(s) \leq m$, every ambient- n walk $\text{root}_n \rightsquigarrow s$ of grade g can be replaced, at grade no greater than g , by the root tower $\text{root}_n \rightsquigarrow \text{root}_m$ of grade $C(n) - C(m)$ followed by a walk confined to states of width $\leq m$.

H5 (gated tail). There is a computable $n_{\min}(\ell)$ such that for every $v > n_{\min}(\ell)$ the stratum $W(v, \ell)$ is the single chain state root_v .

H0 and H2 are signature constraints rather than hypotheses, and in the mechanization they become exactly that (Section 4). H1 says the ambient is a ceiling and nothing else: it may open routes into wider states, but it may not alter the width- $\leq m$ part of the transition structure. H3 names the forced prefix and may be trivial.

H4 is where the mathematics is. It says that no detour through high-width states is ever a shortcut. Where width can only decrease along an edge, H4 is free and therefore invisible: our own first statement of the theorem, phrased for sorting networks, did not contain it, because there Succ preserves width and Prune lowers it by one, so every geodesic descends monotonically. The moment an edge can *raise* the width, a large-ambient search acquires routes unavailable at a smaller ambient that nevertheless end inside the width- $\leq m$ stratum, and H4 becomes the genuine claim that those routes never help. That is why the second domain of Section 6 was chosen to be one in which width is neither monotone up nor down: H4 is the hypothesis that can fail, and it deserved an adversarial test.

The form of H4 printed above is a *decomposition*: tower, then confinement. It is not the form we first published, and the difference is not cosmetic; see Section 4.4.

H5 is strong and rare, and is stated above at the small ambient v itself rather than as a property of the large search: weaker as a hypothesis, stronger as a theorem. It forces $w(\text{root}_v) = v$ above the threshold, so it cannot hold with a degenerate root tower unless the strata are empty.

2.2 Which half needs what

The hypotheses split the theorem in two, and the split is the main structural finding of the abstraction.

proof step	H0	H1	H2	H3	H4	H5
geodesic confinable to width $\leq m$					yes	
initial segment is the root tower, grade $C(n) - C(m)$				yes		
remaining edges exist at ambient m		yes				
grades agree			yes			
states carry the same names at both ambients	yes					
converse direction		yes	yes	yes		
the tail above $n_{\min}(\ell)$ is a chain				yes		yes

Part I (ambient collapse) needs H0–H4; Part II (chain collapse) additionally needs H5 *and* needs H3 to be non-degenerate. The useful statement is not that Part I is independent of H3 — it is not — but that Part I holds with a *degenerate* root tower, in which case H3 is vacuous. The split is therefore between the hypotheses that make the tail trivial (H3-non-degenerate together with H5) and everything else.

3 The theorem and the growth law

Theorem 2 (Ambient Collapse). *Assume H0–H4. Then for all $m \leq n$ and every level ℓ ,*

$$\text{Reach}(n, \ell) \cap \{s : w(s) \leq m\} = \text{Reach}(m, \ell),$$

and $\text{level}_n(s) = \text{level}_m(s)$ for every s of width $\leq m$, where $\text{level}_n(s)$ is the least ℓ with $s \in \text{Reach}(n, \ell)$.

Proof sketch. Fix s with $w(s) \leq m \leq n$ and an ambient- n walk from root_n to s of grade $g \leq C(n) + \ell$. By H4 replace it, at no greater grade, by the root tower down to root_m — of grade exactly $C(n) - C(m)$ by H3 — followed by a walk confined to width $\leq m$ of grade g' with $(C(n) - C(m)) + g' \leq g$. Every edge of the confined part joins states of width $\leq m$, so by H1 it is an edge of $\rightarrow_m$, by H2 carries the same grade, and by H0 its endpoints have the same names at both ambients; hence $g' \leq C(m) + \ell$. Conversely, every edge of a walk at ambient m joins states of width $\leq m$, so by H1 read in the other direction it is an edge of $\rightarrow_n$, and prepending the root tower gives an ambient- n walk of grade $C(n) - C(m) + g'$. Equality of levels follows, hence of the width- $\leq m$ parts. $\square$

Corollary 3 (stratification). *Under H0–H4, $\text{Reach}(n, \ell) = \bigsqcup_{v \leq n} W(v, \ell)$, where each stratum is a function of (v, ℓ) alone and carries no dependence on n .*

This is the form a referee should be shown. Theorem 2 is an identity between two sets that both mention n ; Corollary 3 says the ambient has been eliminated — the census is a running sum of n -free pieces — and it follows in one line by applying Theorem 2 at $m := w(s)$.

Theorem 4 (Chain Collapse). *Assume H0–H5, with H3 non-degenerate. Then for all $n \geq n_{\min}(\ell)$,*

$$\text{Reach}(n, \ell) = \text{Reach}(n_{\min}(\ell), \ell) \uplus \{\text{root}_v : n_{\min}(\ell) < v \leq n\},$$

and if $\text{Reach}(n_{\min}(\ell), \ell)$ is finite then $|\text{Reach}(n, \ell)| = |\text{Reach}(n_{\min}(\ell), \ell)| + (n - n_{\min}(\ell))$.

Theorem 4 is the cutoff proper: the census at every ambient follows from one computation at $n_{\min}(\ell)$, and the correction term is an explicit closed form rather than an error bound.

3.1 Three tail regimes

H5 is the strongest hypothesis in the list, and the interesting question is what happens without it. Part I alone forces the census to be a running sum of n -free strata, so $|\text{Reach}(n, \ell)|$ for large n is determined entirely by the shape of $W(\cdot, \ell)$ above the population front. Across the three domains below, exactly three regimes occur.

regime	strata above the front	$ \text{Reach}(n, \ell) $, large n	instance
T-empty	nothing	constant	Boolean chains
T-trivial (= H5)	one state per width	$\text{const} + n$	sorting networks
T-poly	a polynomial in w	polynomial in n	prefix reversal

Corollary 5 (census growth law). *Assume H0–H4, that every stratum is finite, and that $W(\cdot, \ell)$ agrees with a polynomial of degree $\leq d$ for all $v \geq v_0$. Then $|\text{Reach}(n, \ell)|$ agrees, for $n \geq v_0$, with a polynomial in n of degree $\leq d + 1$; consequently one run at ambient $v_0 + d + 1$ determines the census at every larger ambient exactly.*

Corollary 5 is what makes the method survive the loss of H5, and in practice T-poly is the regime it most often lands in.

3.2 Affine, not polynomial: a distinction we must not blur

Three different statements travel under the phrase “polynomial growth law” and they have three different statuses. We separate them once and keep them separate.

Growth in the ambient, gated case (PROVEN). Under H0–H5 the census is eventually affine in n with slope exactly 1: Theorem 4, machine-checked. Degree one is a polynomial, so “polynomial growth law” is not false, but it oversells and obscures the sharp part; what distinguishes this from [29] — exact equality with no growth term at all — is fully carried by “affine with slope exactly one”. For the sorting-network search we have checked it against every census figure in the artifact: at levels 1, 2, 3, 4 the census over ambients 6–13, 7–13, 9–13 and 9–13 is an arithmetic progression of common difference exactly 1, and at levels 1–5 the width census at ambient 13 holds exactly one state at every width strictly above $n_{\min}(\ell)$.

Growth in the ambient, polynomial case (CONDITIONAL; machine-checked as an implication). Corollary 5 is a mechanized theorem, but its hypothesis — that the strata are eventually polynomial — is a measured fact in the one domain where it applies, not a derived one, and there we prove only the matching upper bound (Section 6). “A polynomial growth law for the census” is defensible only as this conditional, and only for that domain.

Growth in the level (NOT A LAW). For the sorting-network search $|\text{Reach}(n_{\min}(\ell), \ell)|$ at $\ell = 1, \dots, 6$ is

$$33, \quad 445, \quad 24,199, \quad 207,995, \quad 50,602,869, \quad 95,221,143,$$

with successive ratios 13.48, 54.38, 8.60, 243.29, 1.88 — a spread of $241\times$. No forward difference of orders one through five vanishes anywhere in the measured range. The alternation is structural: the ratio is large exactly when $n_{\min}$ jumps and small when it does not. Nothing in this paper bounds growth in the level, and any sentence about polynomial growth of the census must say *in the ambient*.

4 The mechanization

The abstract theorem is machine-checked in Isabelle/HOL [32]. The development is one file, `Ambient_Collapse.thy`, 1,095 lines, built under Isabelle2020 in its own session in about two seconds on top of a prebuilt HOL heap. It contains no `sorry`, no `oops`, no `axiomatization` and

no `quick_and_dirty`; the build script greps for all four and refuses to invoke Isabelle if any is present. The file `imports Main` and nothing else — in particular it never touches the verified checker theories of Section 7, so the two formal artifacts are independent.

4.1 The locale

Walks are formalised by an inductive predicate `pathg E G s t g` — a walk from s to t of grade exactly g — and a variant `pathgB` carrying a predicate every intermediate state must satisfy, used for width confinement. The census is defined by an *inequality* on the grade,

`Reach n l = {s. EX g. pathg (edge n) grade (root n) s g & g <= C n + l}`

which is definitionally the ball $\{s : d_n(\text{root}_n, s) \leq C(n) + \ell\}$ whenever the distance exists and is total when it does not, so the weighted distance never has to be constructed and no partiality enters.

H0 and H2 are discharged *structurally* and cannot be violated by any interpretation: H0 is the fact that states inhabit a single type `'s` not mentioning n , so a state has one name at every ambient by construction, and H2 is the fact that `grade` has type `'s  $\Rightarrow$  's  $\Rightarrow$  nat`. These are signature constraints rather than hypotheses, and they are the honest formal content of “canonicalisation quotients away the unused dimensions”. The rest are locale assumptions:

```
locale graded_ambient_search =
  fixes width :: "'s => nat" and edge :: "nat => 's => 's => bool"
    and grade :: "'s => 's => nat" and root :: "nat => 's"
    and C :: "nat => nat"
  assumes H1: "[| width s <= m; m <= n |] ==>
    {t. edge n s t & width t <= m} = {t. edge m s t}"
    and H3_width: "width (root m) <= m"
    and H3_mono: "m <= n ==> C m <= C n"
    and H3_tower: "m <= n ==> pathg (edge n) grade (root n) (root m) (C n - C m)"
    and H4: "[| m <= n; width s <= m; pathg (edge n) grade (root n) s g |] ==>
      EX g'. pathgB (edge n) grade (%t. width t <= m) (root m) s g'
        & (C n - C m) + g' <= g"

locale gated_ambient_search = graded_ambient_search +
  fixes nmin :: "nat => nat"
  assumes H5: "nmin l < v ==> stratum v l = {root v}"
```

H1 is transcribed verbatim as a set equality; the mechanization extracts its three consequences separately, one of which — that an ambient- m edge out of a width- $\leq m$ state lands in width $\leq m$ — is implied by the equality but was never stated in our prose, and is used twice.

4.2 What is proved

The following are theorems of the locale, stated here exactly as they appear in the file.

```
theorem ambient_collapse: "m <= n ==> Reach n l Int {s. width s <= m} = Reach m l"
corollary level_agree: "[| m <= n; width s <= m |] ==> level n s = level m s"
theorem census_stratified: "Reach n l = (UN v<=n. stratum v l)"
theorem chain_collapse: "nmin l <= n ==>
  Reach n l = Reach (nmin l) l
  Un root ' {v. nmin l < v & v <= n}"
theorem chain_collapse_card: "[| nmin l <= n; finite (Reach (nmin l) l) |] ==>
  card (Reach n l)
    = card (Reach (nmin l) l) + (n - nmin l)"
theorem census_affine: "[| !!v. finite (stratum v l); nmin l <= n |] ==>
  fdiff (%n. census n l) n = 1"
(* stratum v l = {s : Reach v l. width s = v}; fdiff f n = f (Suc n) - f n *)
```

Corollary 5 is mechanized as `census_growth_law` in vanishing-forward-difference form, then converted by a supporting lemma `newton_expansion` — vanishing $(d+1)$ -st differences imply

agreement with $\sum_{j \leq d+1} c_j \binom{t}{j}$ — proved constructively by telescoping and the hockey-stick identity. Two reasons for the difference form. First, vanishing forward differences are what the diagnostic tooling measures, so the mechanized statement and the measured statement coincide. Second, it makes `census_affine` literally the $d = 0$ case: T-trivial and T-poly are one theorem at two degrees.

4.3 Non-vacuity: two interpretations

Locale theorems are worthless if the assumptions are contradictory, so both locales are interpreted. Neither interpretation formalizes a real domain; both are the smallest honest models that exercise the hypotheses, and both discharge every obligation.

Words — states are `nat list`, the ambient is the alphabet size, the width of a word is the least alphabet containing it, and the two moves, each of grade one, are appending a letter $a < n$ and dropping the last letter. The tower is degenerate. The mechanization checks that `Reach 2 1` $\subsetneq$ `Reach 3 1` — so the ambient really matters — and that there are edges which *raise* the width and edges which *lower* it. H4 therefore has genuine content here and is proved, not assumed away: because every move changes the length by exactly one and costs exactly one, a word of length k cannot be reached in fewer than k moves however far the walk climbs in between. That is the “no high-width detour is a shortcut” statement, discharged.

Tower/Gated — a two-constructor datatype whose chain $\text{Tow}(n) \rightarrow \text{Tow}(n-1) \rightarrow \dots$ has grades $\delta(k)$, giving a non-degenerate free chain, and in which each $\text{Tow}(k)$ has exactly one off-chain successor of the *same* width, reachable only at grade $C(k) + D(k) + 1$. With D monotone and unbounded, H5 holds; the cardinality identity and `fdiff`($\lambda n. \text{census } n \ell$) = 1 above the cutoff are both checked. This is the abstract skeleton of the sorting-network instance and the honest bridge for the scope gap of Section 5: it reproduces the bound gate by *pricing* the off-chain edge, turning a level-dependent control-flow test into a level-independent grade. It is not a formalization of that search: here the gate is postulated, and in the real engine it is derived.

The *Words* interpretation also earns its keep numerically. Its width- v stratum at level ℓ is $\sum_{k \leq \ell} (v^k - (v-1)^k)$, of degree $\ell - 1$ with leading coefficient exactly ℓ , and its sphere at distance ℓ has size n^ℓ . Those are precisely the degree and leading coefficient the prefix-reversal domain exhibits (Section 6) — produced by a model with no prefix-reversal content whatsoever. We draw the deflationary conclusion in place rather than leave it to a referee.

4.4 Three corrections the mechanization forced

Mechanizing a theorem one believes is a way of finding out what one actually believes. Three of our own published hypotheses did not survive.

(1) *Part I needs H3*. We had asserted that Part I uses neither H3 nor H5. That is false, and it contradicted our own step table. Part I relates the ambient- n distance from root_n to the ambient- m distance from root_m ; without a hypothesis relating the two roots and pricing the passage between them, the distances are incomparable. In the mechanization H3 appears twice: as `H3_tower` in the converse direction, and inside the statement of H4 in the forward direction, where the $C(n) - C(m)$ term is what accounts for the tower prefix. What survives is the statement at the end of Section 2: Part I is independent of H5 and holds with a *degenerate* root tower.

(2) *H4 as originally printed is ill-formed*. Our published H4 read: for every s with $w(s) \leq m$ and every $n \geq m$, some γ -geodesic $\text{root}_n \rightsquigarrow s$ passes only through states of width $\leq \max(m, w(\text{root}_m))$. When the tower is non-degenerate, $w(\text{root}_n) = n > m$ and root_n lies on every geodesic out of itself, so the condition is unsatisfiable in exactly the case in which it is interesting — which is the sorting-network case. It is satisfiable in the two degenerate-tower domains, which is why our Python hypothesis-checkers never caught it. The repair is the

decomposition form of Section 2, fusing the first two steps of the informal proof into one hypothesis; the max form is not repairable by adjusting the bound.

Being candid about what this costs: H4-as-decomposition now carries the mathematical weight of Part I. The mechanization proves the *transport* (a width-confined ambient- n walk is an ambient- m walk of the same grade, and conversely) and the *bookkeeping* (the levels line up under $C(n) - C(m)$). It does not prove H4 for any real domain. In the two interpretations H4 is discharged, non-trivially in **Words**; in sorting networks it is asserted to follow from width closure, and in prefix reversal it is verified over a finite range, not proved.

(3) *H5 belongs at the small ambient.* We had stated H5 as a property of the stratum inside the large search. Stating it at ambient v itself and letting Part I lift it is weaker as a hypothesis and stronger as a theorem, and it makes the mechanized Part II a derivation rather than a restatement. It also lets us *derive* what we had only observed: the only domain with a non-degenerate root tower is the only domain with a trivial tail.

4.5 What is not mechanized

Stated flatly, because the distinction is the entire value of the section.

1. **That the sorting-network search is an instance is not mechanized.** It rests on an exhaustive reading of the engine’s recursion, plus a second reviewer’s audit (Section 5). What changed with the mechanization is the *shape* of the residual risk: before, an abstract theorem proved on paper resting on an unverified implementation property; now, a machine-checked theorem plus an unverified claim that one concrete search satisfies five stated properties. The second obligation is smaller and far better localised. We do not say the collapse theorem “is verified for sorting networks”.
2. **The class definition does not literally model that engine.** The census of Definition 1 is a ball; the engine stores a bound-gated subset of one. Section 5 gives the level-1 disagreement explicitly.
3. **The two further domains are not formalized.** They are checked by a Python harness, with a brute-force validator for one of them; formalizing the prefix-reversal domain would require *proving* H4 for it, which is open.
4. **Finiteness is an explicit hypothesis, not a consequence.** The three cardinality theorems all assume the strata finite; nothing in H0–H5 implies it, and it is discharged in both interpretations.

5 The sorting-network instance

The concrete search is Harder’s size-optimality dynamic program for sorting networks [24], the engine that settled $S(11) = 35$ and $S(12) = 39$; it is the size-optimality counterpart of the SAT- and symmetry-break-driven searches that settled the smaller cases [11, 12, 8, 17]. A state is a canonical output set of width k ; the two moves are applying a comparator (Succ, width-preserving) and conditioning on an extremal channel and deleting it (Prune, width-lowering by one); the bound oracle combines a Huffman-style rule [26] over the pruned children with successor expansion. The ambient is the channel count.

The semantic half of the instantiation is *not* a mathematical novelty, and we say so first. Harder defines the objective for a sequence set with no reference to any ambient, so “a bound on X is valid at every ambient $n \geq w(X)$ ” is immediate from his definition. What is ours is an *implementation audit*: the claim that the engine actually respects that, established by reading every loop bound, table index and ordering key inside the recursion and confirming each is the

current state’s own channel count or a stored bound; a second reviewer given the same code as a refutation target; and the identification of the single place where it fails.

That place is the on-line subsumption index, whose default width policy is keyed to the *root* width, at widths $\{n - 3, n - 2\}$ — different absolute widths at every ambient — and which is read inside the recursion. With it enabled under that policy, H1 is false. The repair is one line of configuration, not of code: pin the policy to absolute widths. The collapse check is what found it, and an identical synthetic leak in a control domain (Section 6) breaks the collapse immediately and visibly.

Given the audit, the instance rests on three facts about the root, each machine-checked against dumps of the engine’s stored states. Every channel of the full cube is extremal for both polarities, so $\text{Prune}(\text{cube}_k)$ is the single state cube_{k-1} and the Huffman rule at cube_k combines k identical children: the free chain is the chain of full cubes and $C(k) = C(k - 1) + \lceil \log_2 k \rceil$. The symmetric group acts transitively on unordered pairs and fixes cube_k setwise, so $\text{Succ}(\text{cube}_k)$ is a single state. The Huffman ceiling at cube_k is $S(k - 1) + \lceil \log_2 k \rceil = C(k) + D(k - 1)$ where $D(n) = S(n) - C(n)$, and successor expansion at the root is not reached below that value; hence H5 holds with $n_{\min}(\ell) = \min\{n : D(n) \geq \ell\}$ and the tail is one state per width. Note that $C(n) = \sum_{k \leq n} \lceil \log_2 k \rceil$ is Knuth’s binary-insertion number [30]: classical, and what is ours is only its identification as the cost of a forced prefix and the reparameterisation $\ell = L - C(n)$.

The scope gap, stated plainly. The engine’s census is not the ball of Definition 1. Grade is one per comparator, so the unique width-13 successor of cube_{13} lies at distance 1 and belongs to the ball at level 1; but the measured level-1 census at ambient 13 contains exactly one state of width 13, namely cube_{13} itself. Ball and census disagree at level 1. The reason is the gate: the recursion reaches successor expansion only when the Huffman test passes, and that test depends on the level, which Definition 1 forbids. Consequently *the abstract theorem is a guide here, not a proof*. The sorting-network identity is proved directly from the gate, and must not be presented as a corollary of Theorem 4.

The prediction, and the folklore objection. From the machine-checked table $D(3 \dots 12) = 0, 0, 1, 1, 2, 2, 4, 4, 6, 6$ we get $n_{\min}(\ell) = 5, 7, 9, 9, 11, 11, 13$ for $\ell = 1, \dots, 7$. This was derived before the censuses were computed and then checked against them: the population front of the ambient-13 level- ℓ census sits at width 5, 7, 9 for $\ell = 1, 2, 3$, and every width above the front holds exactly one state, at lower bound $C(w) + \ell$. At level 5 — 50,602,871 states — the front is at width 11, and widths 12 and 13 hold one state each.

The bounds these censuses bear on are also the ones circulating in the public tables of sorting-network sizes [16], and lower bounds for restricted classes remain an active target [33, 35, 21]. The deflationary explanation for any census stabilising in n is the folklore support bound: an object built from L operations touches at most $2L$ coordinates, so the census cannot change once $n \geq 2L$. Here $L = C(n) + \ell$ with $C(n) = \Theta(n \log n)$, so the support bound demands $n \geq 2(C(n) + \ell)$, which is false for every $n \geq 3$ and becomes more false as n grows.

level ℓ	$n_{\min}(\ell)$	comparators $L = C(n_{\min}) + \ell$	support bound $2L$	$2L \leq n_{\min}$?
1	5	9	18	no (3.6× over)
2	7	16	32	no (4.6× over)
3	9	24	48	no (5.3× over)
4	9	25	50	no (5.6× over)
5	11	34	68	no (6.2× over)
6	11	35	70	no (6.4× over)

The support bound never certifies stabilisation at any ambient the search can reach, at any level, and the gap widens with n . The cutoff here comes from the gated root tower, which is a

different mechanism.

6 Two further domains

A method claim needs instances that were not the source of the abstraction. We ran two under a pre-registration protocol: compute the small instance, derive the prediction *from that file alone*, hash it, record the hash, and only then compute the larger ambients. The check is on the SHA-256 of the sorted key list, not on counts, so a collapse holding numerically but not state by state would fail.

Boolean chains — a positive control. A straight-line program over $\{\wedge, \vee, \oplus\}$ starts from n projections and adds gates; the state is the multiset of gate outputs, the level is the gate count, canonicalisation drops variables outside the essential support of every computed function, and the width is the number of essential variables. Width is *non-decreasing* along an edge — the opposite of the sorting-network case, so Part I’s proof runs in the other direction — and the root has width 0, so H3 is degenerate. Each gate introduces at most two fresh variables, giving $n_{\min}(\ell) = 2\ell$ with an *empty* tail. Every pre-registered prediction held key for key: at level 3, ambients 2–10, all 36 pairs $m < n$ pass with symmetric difference zero and no level disagreements, the census being a step function saturating at 541 from ambient 6; at level 4 the same at 11,263 states from ambient 8. A raw enumeration with no canonicalisation and no modelling shortcut reproduces the search exactly and shows the canonical form to be a *complete* invariant, so the collapse is not an artefact of a too-coarse quotient.

We grade this domain honestly: $n_{\min}(\ell) = 2\ell$ *is* the folklore support bound. It demonstrates that the framework is correct and that Part I holds with a degenerate tower and an upward-monotone width — a positive control, not independent evidence that census cutoffs ever beat folklore.

Prefix reversal — the real test. A pancake flip f_k reverses the first k entries of a permutation of $[n]$ [22, 7]; the level is the flip count, canonicalisation strips the maximal sorted suffix, and the width is what remains. Two properties make this the adversarial case. No support bound exists at all: a single flip f_n produces a state of width exactly n . And width is neither monotone up nor monotone down — a flip past the current width drags a sorted element to the front, and a later flip can bring it back — so a search at ambient n genuinely has routes unavailable at ambient $m < n$ that end inside the width- $\leq m$ stratum. Part I is precisely the claim that no such detour is ever a shortcut, and that was the pre-registered falsification target.

It was not falsified. At level 5, for all 45 ambient pairs $m < n$ in $3 \dots 12$, the width-restricted censuses agree with symmetric difference zero and no level disagreements on any shared key; a separate state-by-state check of H1 against ambient 11 found 0 mismatches over 120, 2,086 and 13,400 states. At level 6, ambients 8, 10, 12, 14 — up to 2,087,574 states — all 6 pairs pass. With the two Boolean levels that is 93 ambient pairs (m, n) with $m < n$ in total and no discrepancy anywhere. **H4 is verified, not proved:** whether a high-width detour can be a shortcut at some larger level is an open proof obligation, and the plausible argument (a flip past the width displaces sorted elements that must all be restored) is not a proof.

Here H5 fails: the tail is 1, 4, 18, 96, 465, 1501, 3687, $\dots$, and it is exactly polynomial. The pre-registered fit — degree $\ell - 1$, leading coefficient exactly ℓ , valid for $w \geq \ell + 1$, taken from widths ≤ 12 only — reproduced widths 13, 14, 15 exactly at levels 3, 4, 5, and extrapolated to a level 6 that had not been computed when the prediction was frozen: predicted $W(13, 6) = 497,108$ from a degree-5 fit at $w = 7 \dots 12$, observed 497,108; sixth forward difference zero; leading coefficient 6. Summing the strata gives a statement about a classical object with no reference to our machinery: *the number of permutations of $[n]$ at prefix-reversal distance exactly ℓ from the identity is a polynomial in n of degree exactly ℓ with leading coefficient exactly 1*. We prove the

upper bound $(n - 1)^\ell$; the matching lower bound is open. Eventual polynomiality of pancake sphere sizes is in any case already known in the permutation-patterns literature, which is the best outcome available for a method claim: the abstract theorem predicted, from hypotheses checked mechanically in seconds, a result proved independently elsewhere. It is a prediction of a known result, not a new one — and the deflationary reading of Section 4 applies, since counting ℓ -letter words over an n -letter alphabet already gives that degree and that leading coefficient.

Two things did fail, and both are recorded. Our threshold formula $n_{\text{stab}}(\ell) = \ell + 1$, sharp for $\ell \leq 6$, does not extend to $\ell = 7$, where the onset is $\ell + 2$; what survives unqualified is that the strata are eventually polynomial of degree $\ell - 1$ and that *some* computable n_{stab} exists, which is all Corollary 5 needs. And a control domain carrying a synthetic pruning heuristic keyed to the root width — the same shape as the sorting-network leak — breaks the collapse at once, with 285 keys present at one ambient and absent at the next. The mathematics of Part I can be true while an implementation detail destroys its applicability; any deployment of the method must run the check rather than assume it.

Read across the two domains: Part I generalizes and Part II does not. *“Chain collapse generalizes” is false. “Ambient collapse generalizes, and chain collapse is the special case in which a gated root tower makes the tail trivial” is true*, and it is the stronger statement, because it says *why* the sorting-network search gets the clean identity. Neither new domain has a non-degenerate root tower, and neither has a gate.

7 A certified census: $S(11) = 35$

The theorem’s payoff is that a census computed once at $n_{\min}(\ell)$ stands for every larger ambient. To exercise that we rebuilt the search from the published description, re-derived $S(11) \geq 35$ from scratch, and had the certificate checked by an unmodified extraction of the verified checker accompanying the original work [24], in the lineage of the extracted certified checkers of Cruz-Filipe and co-authors [15, 14] and resting on Isabelle’s code generator [23]. The point is not the value, which is known: it is that an independently produced certificate is accepted by a checker nobody here wrote, and that the census the run produces is the one the theorem says determines every larger channel count.

The run was in four stages. Search, at ambient 11 with the subsumption index pinned to absolute widths $\{8, 9\}$, took 71 h 06 m at a peak of 13.08 GB on a six-core Ryzen 3600 with 47 GB of memory, and returned the bound 35 with 95,221,143 stored states — reconstructed from the run log as 103,343,397 insertions less 8,122,254 evictions. Prune-all took 9 h 45 m at about 8.4 GB. Certificate assembly took about 9 h and was killed twice by the out-of-memory killer, at an anonymous resident set of 37.9 GB on a machine with no swap; it completed on the third attempt only after a 96 GB swap file was armed underneath the live process, so its true peak is unknown and is by construction above 37.9 GB. It emitted a certificate of 2,442,317,348 bytes (2,329 MiB), SHA-256 672c433f..., containing 10,275,769 steps. Verification, on an Apple M4, took 89 m 24 s at a peak of 4.35 GB and returned **Just (11, 35)** with exit status 0. From the machine-checked correctness proof of the certificate-checking routine we therefore get $S(11) \geq 35$, matching the known upper bound.

stage	original [24]	this work	memory	wall
search	178 GiB (191 GB), 4 h 51 m	13.08 GB, 71 h 06 m	14.6× less	14.7× more
prune-all	16 GiB, ≈2 d 5 h	~8.4 GB, 9 h 45 m	2.1× less	5.4× less
gen-proof	54 GiB (58.0 GB), 19 h 02 m	> 37.9 GB, ~9 h	1.5× less	2.1× less
verify	6 GiB, 34 m	4.35 GB, 89 m	1.5× less	2.6× more
certificate	2,926 MiB	2,329 MiB	1.26× smaller	—
total wall	77.5 h	91.3 h	—	1.18× more

Three things about that table have to be said rather than inferred. First, the $14.6\times$ is a *search-stage* figure and nothing more; an earlier version circulated as $13.6\times$, obtained by dividing 178 GiB by 13.08 GB as though the units matched, and the corrected ratio is $191.13/13.08 = 14.61$. Second, the memory was bought with time on weaker hardware: the same stage is $14.7\times$ slower and end to end the pipeline is $1.18\times$ slower — a memory-for-time trade, not an improvement. Third, the binding memory peak *moved stages*: in the original it was search, here it is certificate assembly, and the end-to-end improvement is only about $1.5\times$ — optimistic even so, since its denominator is a lower bound from a failed attempt. Certificate generation, not search, is the hidden peak of this pipeline and should be budgeted separately.

We also record what the run does *not* establish, since the census is the object the rest of the paper depends on. Only verification left a surviving machine-readable artifact; the other three stages are recorded in prose and a run log, and no timing output survives that would settle whether 13.08 is gigabytes or gibibytes. The wall-time column compares three machines and two configurations and is not a controlled measurement. And the census figure that Theorem 4 exports — $|\text{Reach}(13, 6)| = 95,221,145$ — is a *prediction* from the single ambient-11 measurement, not a measurement.

8 Where the method stops

A cutoff theorem tells you which computation to run. It does not tell you that you can afford it, and in this programme the two answers diverge at the next level. We report the divergence because it delimits the method.

Theorem 4 makes the level-6 census at ambient 13 a corollary of the ambient-11 run of Section 7: $n_{\min}(6) = 11$, so no computation at 13 is needed. At level 7 the leverage vanishes, because $n_{\min}(7) = 13$. There is no smaller ambient to collapse to; the theorem is silent, and the cost is whatever the search costs. That is the method’s boundary in the sharpest form available. A cutoff at $n_{\min}(\ell)$ removes the ambient as a cost driver and leaves the level as the only one — and Section 3 showed that census growth in the level is governed by no law we have.

What the cost actually tracks. The measurement that matters here is a diagnosis, not a projection, and it corrects an assumption we had been making: extrapolating from state counts assumes cost is proportional to stored states, and it is not. Between levels 5 and 6 of the certified run, stored entries grew $21.8\times$ and the subsumption index population $20.4\times$, but index candidates examined grew $297\times$ and the iteration wall $418\times$; index lookup accounted for 79.4% of user time. Two independent measurements — the level-to-level ladder, and a 25,613-sample trace of the instantaneous state rate inside level 6, which fell $6.32\times$ while the live set grew $10.3\times$ and was still falling in the final hour — agree that wall grows as $N^{1.79}$ to $N^{1.94}$. So: **memory scales with the census; wall clock scales with the index.** The census is the quantity our theorem controls; the wall is not, and it is the faster-growing one. A four-arm sweep of index configurations found a strict Pareto improvement over both previously used settings — $1.49\times$ faster and $4.2\times$ less memory than an unindexed run, $2.02\times$ faster than the certified setting at $1.86\times$ its memory — which moves the operating point *along* that frontier rather than off it. Projecting each arm on its own measured ladder puts a level-7 search at roughly 45–190 days and 1.5 TB resident against a months-scale horizon and 143 GB of memory plus swap: about $17\times$ short, combined, down from a standing bracket of 2.4–195 TB of storage. The gap has narrowed by roughly three orders of magnitude; it has not closed. Every level-7 figure here is an extrapolation — the run was not performed — and the level-6-to-7 state multiplier is the one input for which no cheaper rehearsal exists, precisely because $n_{\min}(7) = 13$.

Why splitting the problem does not help. The obvious response is to decompose: partition the search by comparator prefix, run k independent jobs, and pay $k \cdot (N/k)^e$ instead of N^e .

Across 146 runs — 144 prefix jobs at six prefix depths and two levels, plus two monolithic controls — the lowest peak per-job memory ever observed was $0.990\times$ the monolithic run’s, and that was at depth 1, where $k = 1$ and the split is a no-op. The best non-trivial point is $0.998\times$; by depth 4 the peak per-job memory is $23.6\times$ the monolith’s. Level 7 would need $0.095\times$.

The failure is at the first premise, not in the arithmetic: $k \cdot (N/k)^e$ assumes the split partitions the state space, and prefix cones *cover* it. At level 5, depth 2, one job’s memo held 11,849,694 entries against the entire monolithic run’s 11,629,126 — 1.9% *larger* than the run it was supposed to be one third of. A memo diff between two sibling cones puts the job-private fraction at 36.2% of entries, so 63.8% of the work is shared and is paid k times over. The mechanism is general: an iterative-deepening branch-and-bound never solves an interior subproblem to completion — it raises the root’s bound incrementally, improving each successor only far enough to move the current minimum — whereas each decomposed job must certify its own root to its full value. The parts are strictly harder than the roles those same parts play inside the whole. Granting the pro-decomposition argument in full — every wide state perfectly private, the low-width remainder free — the census supplies a floor: widths ≥ 10 are 56.7% of packed bytes, so even a perfect decomposition with $k \rightarrow \infty$ and zero CPU cost leaves the memo at 0.43 times its monolithic size, still several times over the available memory. This is a clean negative and it ends that line of work.

9 Trusted base

Following Harder [24], we do not claim the components below are free of bugs and do not expect them to be. The goal is narrower: to say which could, if wrong, move a stated claim. The scope is the specific runs reported here, so only bugs triggered during those runs are relevant, and we adopt the de Bruijn criterion [4] as the standard the certified computation is meant to meet rather than claiming correctness directly.

component	trusted?	note
Isabelle/HOL kernel (Isabelle2020)	Yes	Shared with the checker development, so the two formal artifacts are not independent of each other at this level.
Statements in <code>Ambient_Collapse.thy</code>	Yes	The proofs are machine-checked; a mis- <i>stated</i> theorem is not. Two of our hypotheses were mis-stated and the mechanization caught them (Section 4.4); a third would not be.
Proofs in <code>Ambient_Collapse.thy</code>	No	Machine-checked, no sorry/oops/axiomatization .
That the sorting-network search satisfies H0–H5	Yes	Not machine-checked. Rests on an exhaustive reading of the recursion plus a second reviewer given the code as a refutation target; the one leak found is Section 5. This is the largest single trusted item in the paper.
Our rebuilt search engine	No	Its output is a certificate, checked independently.
Prune-all and certificate assembly	No	Same; an error here yields a rejected certificate.
Extracted verified checker core	Yes	An <i>unchanged</i> extraction of the frozen Checker.thy ; the extraction pipeline was reproduced from source under Isabelle2020.
Checker wrapper and decoder	Yes	Patched , and not verified: a large-read fix in the Haskell wrapper was required on the verification host. “Unmodified” scopes to the extracted core, never to the binary.
Census figures at levels 5 and 6	Yes	One measured ambient each, so the affine law there rests on width-stratum evidence rather than an observed common difference; $ \text{Reach}(13, 6) $ is a prediction of Theorem 4.
Certificate step and memory figures	No	No claim depends on them; the step count is a recount over the certificate, superseding an earlier figure derived from a different quantity.
Hypothesis checks in the two further domains	Yes	Python, not formalized. In the Boolean domain a raw enumeration independently checks the canonical form; in prefix reversal there is no such check, and H4 is verified over a finite range, not proved.
GHC, operating system, hardware	Yes	Standard, and unavoidable.

Two dependencies deserve naming before a referee names them. The mechanized theorem and the certified computation are naturally described as two independent formal artifacts; they share the Isabelle kernel and the same released version of it, so they are independent in content but not in toolchain. And that toolchain is the programme’s most fragile asset: the Isabelle2020 installation lives outside the repository, has had to be re-downloaded once already, and the theory sources are committed while the prover that checks them is not pinned. Of the four stages of Section 7, only verification left a machine-readable artifact.

Finally, a deliberate exclusion. Our own abstract theory sits *outside* the trusted base for the certified bound: every line of Sections 2–4 could be wrong and $S(11) \geq 35$ would not move, because that bound is established by a certificate and a checker that know nothing about ambients, levels or censuses. Conversely, the collapse theorem is what licenses the claim that

one ambient-11 census determines the ambient-13 one, and *that* rests on the audit in the fourth row above — the first place a reader who wants to disbelieve us should look.

10 Conclusion

An exhaustive combinatorial-optimality search can have a cutoff in the sense that parameterized verification has understood the word for thirty years: above a computable threshold, the object being computed at parameter n is determined by the object at the threshold, plus a closed-form correction. We have given hypotheses under which this holds for the census — the set of states such a search stores — proved both halves and the growth law in Isabelle/HOL with two interpretations that make them non-vacuous, and instantiated the class in three domains, one of which is a positive control and one of which was chosen because it could have falsified the theorem and did not. The growth law we prove is affine with slope exactly one; the polynomial version is a machine-checked implication whose hypothesis is measured rather than derived. What we have not shown is that any particular engine is an instance: that remains an audit, and it is the obligation a reader should press on. The method’s practical reach ends where the theorem stops supplying a smaller ambient to collapse to — and there, as we measured, the cost is governed by a quantity the theorem does not control.

References

- [1] Simon Außerlechner, Swen Jacobs, and Ayrat Khalimov. Tight cutoffs for guarded protocols with fairness. In *Verification, Model Checking, and Abstract Interpretation (VMCAI 2016)*, volume 9583 of *LNCS*, pages 476–494. Springer, 2016.
- [2] The authors. A 1972 lower-bound argument for sorting networks, and what rests on it. Companion paper, 2026.
- [3] The authors. Artifact: `Ambient_Collapse.thy`, the rebuilt search engine, the certificate, and the verifiers. Deposit; to appear, 2026.
- [4] Henk Barendregt and Freek Wiedijk. The challenge of computer mathematics. *Philosophical Transactions of the Royal Society A*, 363(1835):2351–2375, 2005.
- [5] Shreesha Bhat and Kartik Nagar. Automating cutoff-based verification of distributed protocols. In *Formal Methods in Computer-Aided Design (FMCAD 2023)*, pages 75–85, 2023.
- [6] Roderick Bloem, Swen Jacobs, Ayrat Khalimov, Igor Konnov, Sasha Rubin, Helmut Veith, and Josef Widder. *Decidability of Parameterized Verification*. Morgan & Claypool, 2015.
- [7] Laurent Bulteau, Guillaume Fertin, and Irena Rusu. Pancake flipping is hard. *Journal of Computer and System Sciences*, 81(8):1556–1574, 2015.
- [8] Daniel Bundala and Jakub Závodný. Optimal sorting networks. In *Language and Automata Theory and Applications (LATA 2014)*, volume 8370 of *LNCS*, pages 236–247. Springer, 2014.
- [9] Joshua Buresh-Oppenheim, Sashka Davis, and Russell Impagliazzo. A stronger model of dynamic programming algorithms. *Algorithmica*, 60(4):938–968, 2011.
- [10] Edmund M. Clarke, Reinhard Enders, Thomas Filkorn, and Somesh Jha. Exploiting symmetry in temporal logic model checking. *Formal Methods in System Design*, 9(1–2):77–104, 1996.
- [11] Michael Codish, Luís Cruz-Filipe, Michael Frank, and Peter Schneider-Kamp. Sorting nine inputs requires twenty-five comparisons. *Journal of Computer and System Sciences*, 82(3):551–563, 2016. Conference version: ICTAI 2014, 186–193; arXiv:1405.5754.
- [12] Michael Codish, Luís Cruz-Filipe, and Peter Schneider-Kamp. Sorting networks: The end game. In *Language and Automata Theory and Applications (LATA 2015)*, volume 8977 of *LNCS*, pages 664–675. Springer, 2015.
- [13] Vianney Coppé, Xavier Gillard, and Pierre Schaus. Branch-and-bound with barrier: Dominance and suboptimality detection for dd-based branch-and-bound. *INFORMS Journal on Computing*, 36(6), 2024.
- [14] Luís Cruz-Filipe, Kim S. Larsen, and Peter Schneider-Kamp. Formally proving size optimality of sorting networks. *Journal of Automated Reasoning*, 59:425–454, 2017.
- [15] Luís Cruz-Filipe and Peter Schneider-Kamp. Formalizing size-optimal sorting networks: Extracting a certified proof checker. In *Interactive Theorem Proving (ITP 2015)*, volume 9236 of *LNCS*, pages 154–169. Springer, 2015.

- [16] Bert Dobbelaere. Smallest and fastest sorting networks for a given number of inputs. https://bertdobbelaere.github.io/sorting_networks.html, 2025. Changelog entry of 2025-04-21; consulted 2026-08-22.
- [17] Thorsten Ehlers and Mike Müller. Faster sorting networks for 17, 19 and 20 inputs. In *arXiv:1410.2736*, 2014.
- [18] E. Allen Emerson and Vineet Kahlon. Reducing model checking of the many to the few. In *Automated Deduction (CADE-17)*, volume 1831 of *LNAI*, pages 236–254. Springer, 2000.
- [19] E. Allen Emerson and Kedar S. Namjoshi. Reasoning about rings. In *Principles of Programming Languages (POPL 1995)*, pages 85–94. ACM, 1995.
- [20] E. Allen Emerson and Kedar S. Namjoshi. On reasoning about rings. *International Journal of Foundations of Computer Science*, 14(4):527–550, 2003.
- [21] Robert W. Floyd and Donald E. Knuth. The Bose–Nelson sorting problem. In *A Survey of Combinatorial Theory*, pages 163–172. North-Holland, 1973.
- [22] William H. Gates and Christos H. Papadimitriou. Bounds for sorting by prefix reversal. *Discrete Mathematics*, 27(1):47–57, 1979.
- [23] Florian Haftmann and Tobias Nipkow. Code generation via higher-order rewrite systems. In *Functional and Logic Programming (FLOPS 2010)*, volume 6009 of *LNCS*, pages 103–117. Springer, 2010.
- [24] Jannis Harder. An answer to the Bose–Nelson sorting problem for eleven and twelve channels. arXiv:2012.04400v3, 2022. Artifacts: `jix/sortnetopt`, doi:10.5281/zenodo.4139152; certificate, doi:10.5281/zenodo.4108365.
- [25] Florian Horn and Arnaud Sangnier. Deciding the existence of cut-off in parameterized rendez-vous networks. In *Concurrency Theory (CONCUR 2020)*, volume 171 of *LIPICs*, pages 46:1–46:16, 2020.
- [26] David A. Huffman. A method for the construction of minimum-redundancy codes. *Proceedings of the IRE*, 40(9):1098–1101, 1952.
- [27] C. Norris Ip and David L. Dill. Better verification through symmetry. *Formal Methods in System Design*, 9(1–2):41–75, 1996.
- [28] Nouraldin Jaber, Swen Jacobs, Christopher Wagner, Milind Kulkarni, and Roopsha Samanta. Parameterized verification of systems with global synchronization and guards. In *Computer Aided Verification (CAV 2020)*, volume 12224 of *LNCS*, pages 299–323. Springer, 2020.
- [29] Alexander Kaiser, Daniel Kroening, and Thomas Wahl. Dynamic cutoff detection in parameterized concurrent programs. In *Computer Aided Verification (CAV 2010)*, volume 6174 of *LNCS*, pages 645–659. Springer, 2010.
- [30] Donald E. Knuth. *The Art of Computer Programming, Volume 3: Sorting and Searching*. Addison-Wesley, 1973. §5.3.4.
- [31] Kedar S. Namjoshi. Symmetry and completeness in the analysis of parameterized systems. In *Verification, Model Checking, and Abstract Interpretation (VMCAI 2007)*, volume 4349 of *LNCS*, pages 299–313. Springer, 2007.
- [32] Tobias Nipkow, Lawrence C. Paulson, and Markus Wenzel. *Isabelle/HOL: A Proof Assistant for Higher-Order Logic*, volume 2283 of *LNCS*. Springer, 2002.
- [33] Akihiko Tozawa and Kunihiro Sadakane. Odd-even transposition sort is an optimal stable standard sorting network. *Information Processing Letters*, 195:106659, 2026.
- [34] Antti Valmari. Stubborn sets for reduced state space generation. In *Advances in Petri Nets 1990*, volume 483 of *LNCS*, pages 491–515. Springer, 1991.
- [35] David C. Van Voorhis. Toward a lower bound for sorting networks. In Raymond E. Miller and James W. Thatcher, editors, *Complexity of Computer Computations*, pages 119–129. Plenum, 1972.